Trade Shocks and Political Change in Local Labor Markets

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1 Summary

Autor et al. (2013) (hereafter ADH), "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," examine how rising Chinese import competition affected U.S. local labor markets between 1990 and 2007. Their primary research question is whether greater exposure to Chinese imports contributed to declines in manufacturing employment, wages, and labor force participation across U.S. regions. Using commuting zones (CZs) as local labor market units, ADH use a shift-share research design that relates changes in local outcomes to differences in baseline industry composition and subsequent industry-level import growth.

Our project extends this framework to political outcomes. We ask whether CZs more exposed to the China shock experienced different changes in Republican presidential vote margins, and whether these effects appear in the short run, medium run, or long run. To answer this question, we combine the ADH replication package with county-level presidential election returns and historical county-to-CZ crosswalks. In minimally controlled specifications, greater trade exposure is associated with more Republican-leaning presidential margins in later elections, but this association largely disappears after adjusting for baseline local characteristics. However, the interpretation is cautious: shift-share designs rely on nontrivial assumptions about initial industry shares and import-growth shocks, and CZ-level political change may reflect belief updating, turnout and registration, elite political supply, or population composition rather than a single mechanism.

2 Introduction

Our project's research question is does exposure to the China trade shock affect political outcomes in the United States? If so, how do these effects evolve over time? The project examines whether commuting zones that were more exposed to rising Chinese import competition experienced different changes in Republican presidential vote margins than less-exposed commuting zones. The question is not only whether trade exposure mattered politically, but also when its effects appeared. For example, we distinguish short-run effects around the Great Recession, medium-run effects around the 2016 election, and longer-run effects that may reflect later shocks such as COVID-19 and inflation.

Understanding this question is important because globalization may reshape political behavior along with the economy. If local labor-market losses from import competition contributed to changes in voting patterns, then trade policy may have broader consequences for democratic representation, polarization, and public support for protectionism. This may help explain why places hit hardest by manufacturing decline may respond differently to national political messages about trade, immigration, redistribution, and economic nationalism.

The question is also important because it connects local economic shocks to national political outcomes. A trade shock that lowers manufacturing employment may eventually influence turnout, party support, media consumption, and attitudes toward government intervention in affected regions. If these effects are persistent, then the consequences of globalization may extend beyond the original employment losses. Thus, studying these dynamics can help policymakers understand the extent of the effect of trade policy and adjustment assistance on political life.

A key empirical challenge is that identification relies on both the measurement of exposure and the validity of the shift-share design. Principally, the ADH exposure measure may suffer from aggregation and temporal bias. Import exposure is aggregated across broad industries, which may conceal substantial heterogeneity at more disaggregated levels. If variation in exposure is driven by a small subset of industries, such as textiles, furniture, and electronics, estimated effects may reflect industry-specific shocks rather than a general China-shock effect.

There are concerns regarding the validity of the Bartik instrument and initial industry shares. Identification depends not only on Chinese export growth but also on baseline industry composition in 1990. These industry shares are not randomly assigned across commuting zones and may be correlated with unobserved determinants of political outcomes, including automation exposure, unionization, education, demographic structure, or pre-existing political trends. Thus, both the IV strategy and reduced-form exposure design require careful discussion of what variation identifies the estimates.

Additionally, there is potential compositional bias due to migration across commuting zones. If trade-exposed regions experience selective in- or out-migration, observed political changes may reflect shifts in population composition instead of changes in individual political preferences. Measurement issues can also arise from mapping county-level election data into historical CZ boundaries and from sensitivity to alternative spatial definitions of labor markets. Together, these concerns motivate our robustness checks, measurement diagnostics, and cautious interpretation.

3 Data

Our analysis combines four primary datasets to construct commuting-zone-level political and trade exposure measures. We use the Autor et al. (2013) replication package for baseline commuting-zone trade exposure measures and pre-determined labor market controls, including industry composition and demographic structure.

To align modern political data with historical labor markets, we use Ferrara et al. (2024), *Crosswalks for U.S. Counties and Congressional Districts, 1790–2020*, which maps contemporary counties to historical boundaries and provides county centroid coordinates. Our preferred crosswalk uses M5 weights where available and M2 weights as fallback for source counties where M5 weights are undefined or zero.

We aggregate counties into 1990 commuting zones using the U.S. Department of Agriculture Economic Research Service Commuting Zones and Labor Market Areas dataset, consistent with the Tolbert and Sizer (1996) classification and the USDA ERS commuting-zone data (U.S. Department of Agriculture Economic Research Service 2026).

Political outcomes are drawn from Amlani and Algara (2021), *Partisanship & Nationalization in American Elections, 1872–2020*, which provides county-level presidential election returns used to construct Republican vote margins.

We follow the sample design of Autor, Dorn, and Hanson (2013). The unit of analysis is the 1990 commuting zone (CZ) of Tolbert and Sizer (1996), covering 722 CZs over 1990–2000 and 2000–2007 stacked in first differences (1,444 observations). Table 1 reports population-weighted means from ADH's replication package. Two findings stand out. Chinese imports per worker (10-year equivalent) more than doubled, from \$1.14k in 1990–2000 to \$2.63k in 2000–2007, tracking China's WTO accession in 2001. Manufacturing employment as a share of population declined from 12.69% in 1990 to 8.51% in 2007, while SSDI receipt increased over the same period.

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Variable	1990	2000	2007
Δ imports from China per worker (\$k, 10-yr equiv.)	1.14 (0.99)	2.63 (2.01)	—
Manufacturing employment / pop. (%)	12.69 (4.80)	10.51 (4.45)	8.51 (3.60)
Non-manufacturing employment / pop. (%)	57.75 (5.91)	59.16 (5.24)	61.87 (4.95)
Unemployment / pop. (%)	4.80 (0.99)	4.28 (0.93)	4.87 (0.90)
Not in labor force / pop. (%)	24.76 (4.34)	26.05 (4.39)	24.75 (3.70)
SSDI recipients / pop. (%)	1.86 (0.63)	2.75 (1.04)	3.57 (1.41)

Notes: N = 722 commuting zones for each year. Means are weighted by start-of-period CZ population. Standard deviations in parentheses. Variables follow Autor, Dorn & Hanson (2013) Appendix Table 2; all values reproduced from the official replication package.

4 Analysis and Findings

4.1 Identification Strategy and its Plausibility

OLS estimation of the labor-market effect of Chinese import exposure faces an endogeneity problem. Observed U.S. imports reflect both Chinese supply-side and U.S. demand shocks, biasing OLS toward zero. Autor et al. (2013) report an OLS estimate of -0.171 versus an IV estimate of -0.596.

They address this with a Bartik-style instrument, replacing ΔIPW_{uit} with ΔIPW_{oit}, built from same-period Chinese exports to eight other high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland). The identifying assumption is that within-industry import growth shared by these countries reflects China's supply-side improvements, not correlated demand:

$$\Delta IPW_{uit} = \sum_j \left(\frac{L_{ijt}}{L_{ujt}} \right) \cdot \left(\frac{\Delta M_{ucjt}}{L_{it}} \right),$$

$$\Delta IPW_{oit} = \sum_j \left(\frac{L_{ij,t-1}}{L_{uj,t-1}} \right) \cdot \left(\frac{\Delta M_{ocjt}}{L_{i,t-1}} \right).$$

The second-stage equation is estimated as stacked first differences:

$$\Delta L_{itm} = \gamma_t + \beta_1 \Delta IPW_{uit} + X'_{it}\beta_2 + e_{it},$$

where X_{it} contains start-of-period manufacturing share, demographic controls, the routine and offshorability indices of Autor and Dorn (2013), and Census division dummies.

Three pieces of evidence support the strategy. First, the first stage is strong: the instrument predicts the endogenous regressor with coefficient 0.631 ($t = 6.90$), well above the weak-IV threshold. Second, a falsification test shows that 1970s–1980s manufacturing changes do not predict future Chinese exposure, ruling out a pre-existing decline. Third, dropping suspect industries leaves the coefficient near -0.596.

The main residual concern is correlated demand shocks across high-income economies. A gravity-based check yields similar estimates; Chinese manufacturing TFP grew about 8% per year over 1998–2007, against 3.9% in the U.S., supporting the supply-driven interpretation.

4.2 Reproduction of Published Results

We verify that the official replication package functions correctly. Our target is Table 3 of Autor et al. (2013), the 2SLS regression of CZ-level change in manufacturing employment share on ΔIPW_{uit}, instrumented by the eight-country measure. We focus on column (6) with full controls.

Running the official Stata code on the published workfile, we obtain coefficients matching the published values exactly to three decimal places across all six columns. In column (6), the coefficient on ΔIPW is -0.596 (SE 0.099); the first-stage coefficient is 0.631 ($t = 6.90$); $N = 1{,}444$ with $R^2 = 0.343$. Table 2 reports the full comparison.

Table 2: Replication of ADH (2013) Table 3: effect of Chinese import exposure on CZ manufacturing employment share.

Table 2. Replication of ADH (2013) Table 3: Effect of Chinese import exposure on CZ manufacturing employment

	(1)	(2)	(3)	(4)	(5)	**(6)**
Panel A. ADH (2013) published estimates						
Δ imports from China / worker	−0.746	−0.610	−0.538	−0.508	−0.562	−0.596
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
Panel B. Our replication						
Δ imports from China / worker	**−0.746**	**−0.610**	**−0.538**	**−0.508**	**−0.562**	**−0.596**
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
First-stage β (instrument)	0.792	0.664	0.652	0.635	0.638	0.631
	(0.080)	(0.089)	(0.094)	(0.094)	(0.091)	(0.091)
Start-of-period mfg. share		Y	Y	Y	Y	Y
Census division dummies			Y	Y	Y	Y
Demographic controls				Y		Y
Routine / offshorability					Y	Y
Observations	1,444	1,444	1,444	1,444	1,444	1,444
R^2	0.066	0.157	0.282	0.321	0.343	0.343

Notes: 2SLS estimates. Dependent variable is the 10-year-equivalent change in CZ manufacturing employment as a share of working-age population (in percentage points). The endogenous regressor Δ imports from China per worker is instrumented by Chinese imports to eight other high-income countries, weighted by lagged industry-employment shares. Stacked first differences for 1990–2000 and 2000–2007 (N = 722 × 2). Models weighted by start-of-period CZ population share. Standard errors in parentheses, clustered at the state level. Replication coefficients match the published values exactly to three decimal places across all six columns.

Economically, -0.596 implies that a $1,000 per-worker increase in decadal Chinese exposure lowers a CZ's manufacturing share by 0.6 percentage points. ADH attribute 44% of the 1990–2007 manufacturing decline to this channel. Because we use the original data and code, this is a computational reproduction of ADH's published estimates, not a validation of the separate political-outcome analysis. The next section extends the CZ framework to presidential voting using an event study of observed import exposure rather than ADH's original instrumental-variables specification.

4.3 Extension: Trade Exposure and Presidential Voting

We examine the relationship between ADH's measure of Chinese import exposure and presidential voting. The outcome is the Republican two-party presidential vote margin, $(R_{ct} - D_{ct})/(R_{ct} + D_{ct})$, where R_{ct} and D_{ct} denote Republican and Democratic votes. We estimate weighted CZ-level event studies:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{k \neq 1988} \beta_k E_c \mathbf{1}\{t = k\} + X_c' \Gamma_t + \varepsilon_{ct}.$$

Here E_c is cumulative ADH import exposure over 1990–2007 in thousands of dollars per worker; α_c and λ_t are CZ and election-year fixed effects, and $X_c'\Gamma_t$ represents baseline characteristics interacted with election-year indicators. Each β_k measures the difference in the exposure–margin association in election year k relative to the omitted 1988 election; it is not a new trade shock in year k. Estimation uses ADH analysis weights and CZ-clustered standard errors. We compare a minimal specification without interacted controls, core controls (initial manufacturing share, college-educated share, and

foreign-born share), and full controls adding female employment share, routine-task intensity, and offshorability. Interacted controls are standardized for estimation.

County boundaries changed during the election series, whereas ADH measures exposure for 1990 commuting zones. We convert county election returns to 1990 county boundaries using the crosswalks of Ferrara et al. (2024), then assign counties to commuting zones using the definitions of Tolbert and Sizer (1996) distributed by the USDA Economic Research Service (U.S. Department of Agriculture Economic Research Service 2026). We use M5 weights where available and M2 weights where M5 is undefined or zero. We check the share of votes retained and compare estimates using M6, M2, and M4 weights. All four specifications retain the 1990 commuting-zone boundaries; alternative definitions of local labor markets remain a separate question (Foote et al. 2021).

The political event study relates observed ADH exposure to voting outcomes conditional on fixed effects and baseline controls; unlike ADH's original labor-market instrumental-variables analysis, it does not instrument exposure in the main regression. ADH exposure has a shift-share structure, combining initial CZ industry composition with subsequent industry-level import growth. Shift-share identification can rely on exogenous initial industry shares or on exogenous industry-level shocks, but these are distinct assumptions (Goldsmith-Pinkham et al. 2020; Borusyak et al. 2025). Neither is established by the observed-exposure political event study, which also requires assumptions about counterfactual political trends. In our diagnostics, the ADH instrument strongly predicts exposure, but exposure also correlates with baseline manufacturing intensity and routine-task concentration. A strong first stage alone cannot justify causal inference for these political outcomes. Table 3 reports first-stage estimates in the political-outcome sample.

Table 3: First-stage diagnostics for ADH exposure.

Specification	Estimate	Std. error	First-stage F	N
Minimal	0.977	0.085	133.4	722
Core controls	0.812	0.101	64.6	722
Full controls	0.794	0.101	62.0	722

Figure 1 presents the main event-study estimates with confidence intervals based on standard errors clustered by commuting zone. Some alternative standard-error estimates are numerically unreliable; their results appear in Appendix E.

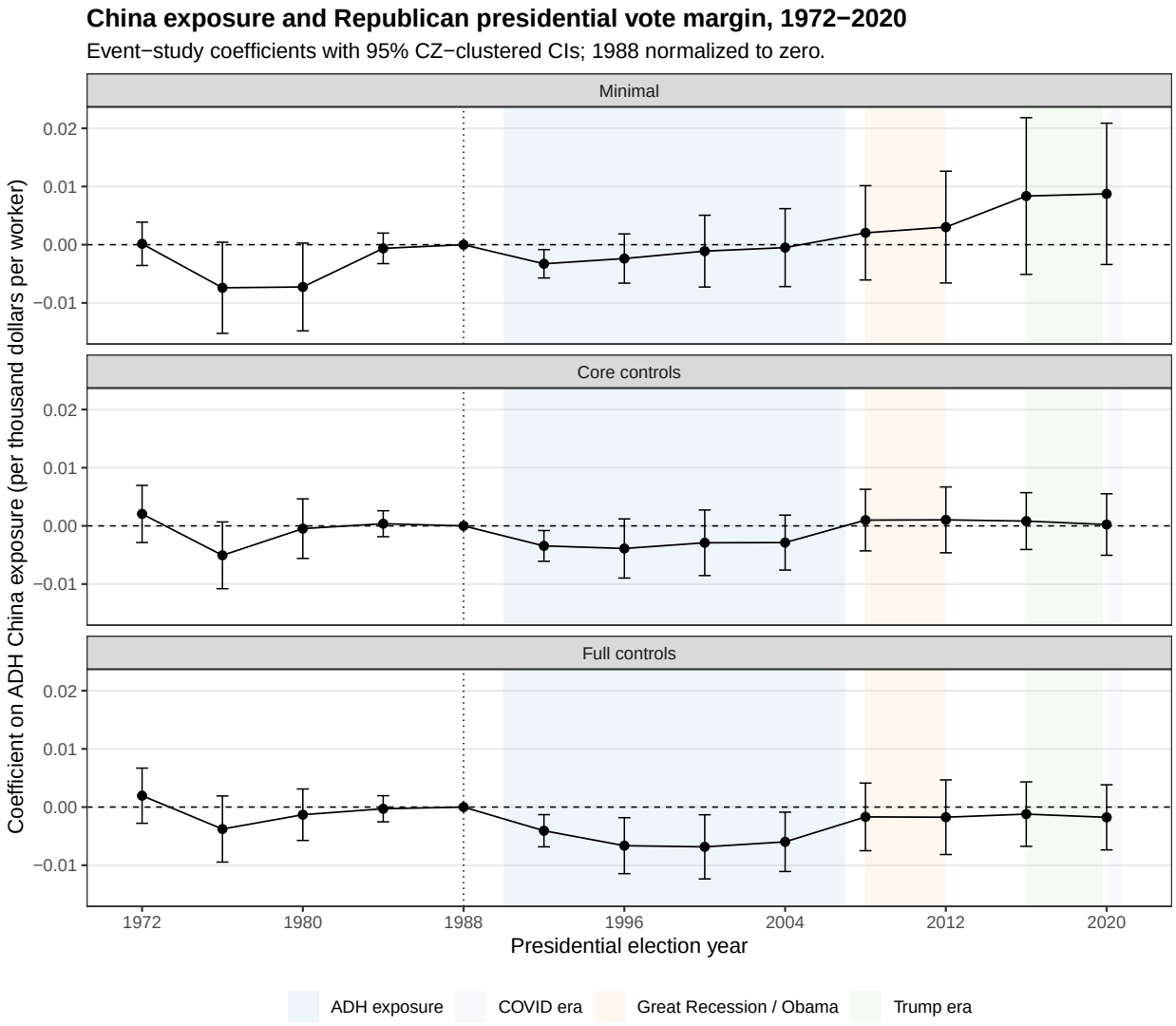


Figure 1: Event-study coefficients for ADH exposure and Republican presidential margin.

The estimates do not reject a joint null of zero pre-1988 exposure coefficients, although this does not establish parallel trends. In 2016 and 2020, the minimally controlled coefficients are positive (0.00837 and 0.00874), but fall to 0.00083 and 0.00024 with core controls and become slightly negative (-0.00121 and -0.00176) with full controls. These coefficients are changes in the Republican two-party margin per additional $1,000 of cumulative exposure per worker, relative to the 1988 association; the 2016 and 2020 full-control estimates are not statistically distinguishable from zero (CZ-clustered $p = 0.668$ and $p = 0.537$). Even the minimally controlled 2016 and 2020 estimates are imprecise. The positive association in the minimal specification largely disappears after adjustment for baseline characteristics. Exposed commuting zones may have differed in their initial economic and demographic characteristics, and political responses may also vary across communities, as Autor et al. (2020) find. Our estimates do not establish which explanation accounts for the differences across specifications.

We also examine Republican and Democratic votes per 1990 population and total two-party votes.

Figure 2 compares these outcomes with the Republican margin. They measure voting totals and electoral composition but do not separately reveal changes in individual preferences, turnout, or migration. Ratios using 1990 population are not turnout rates for later elections.

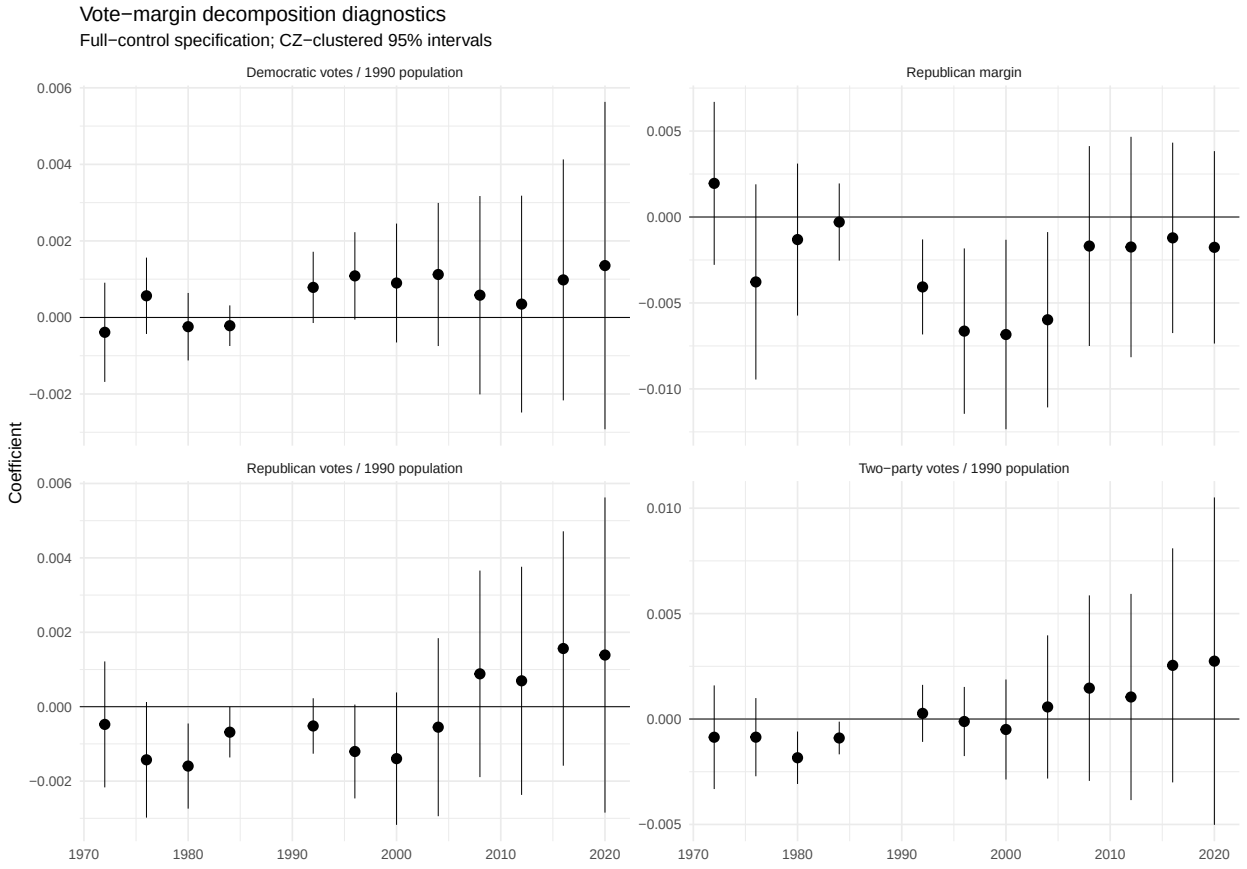


Figure 2: Political-outcome decomposition across election years.

We additionally estimate separate associations for import exposure during 1990–2000 and 2000–2007, entering both measures in the same regression. Figure 3 compares their estimated coefficients across election years. Coefficients before 1988 serve as placebo comparisons with future exposure; those after 2008 describe associations with import growth during the two earlier periods.

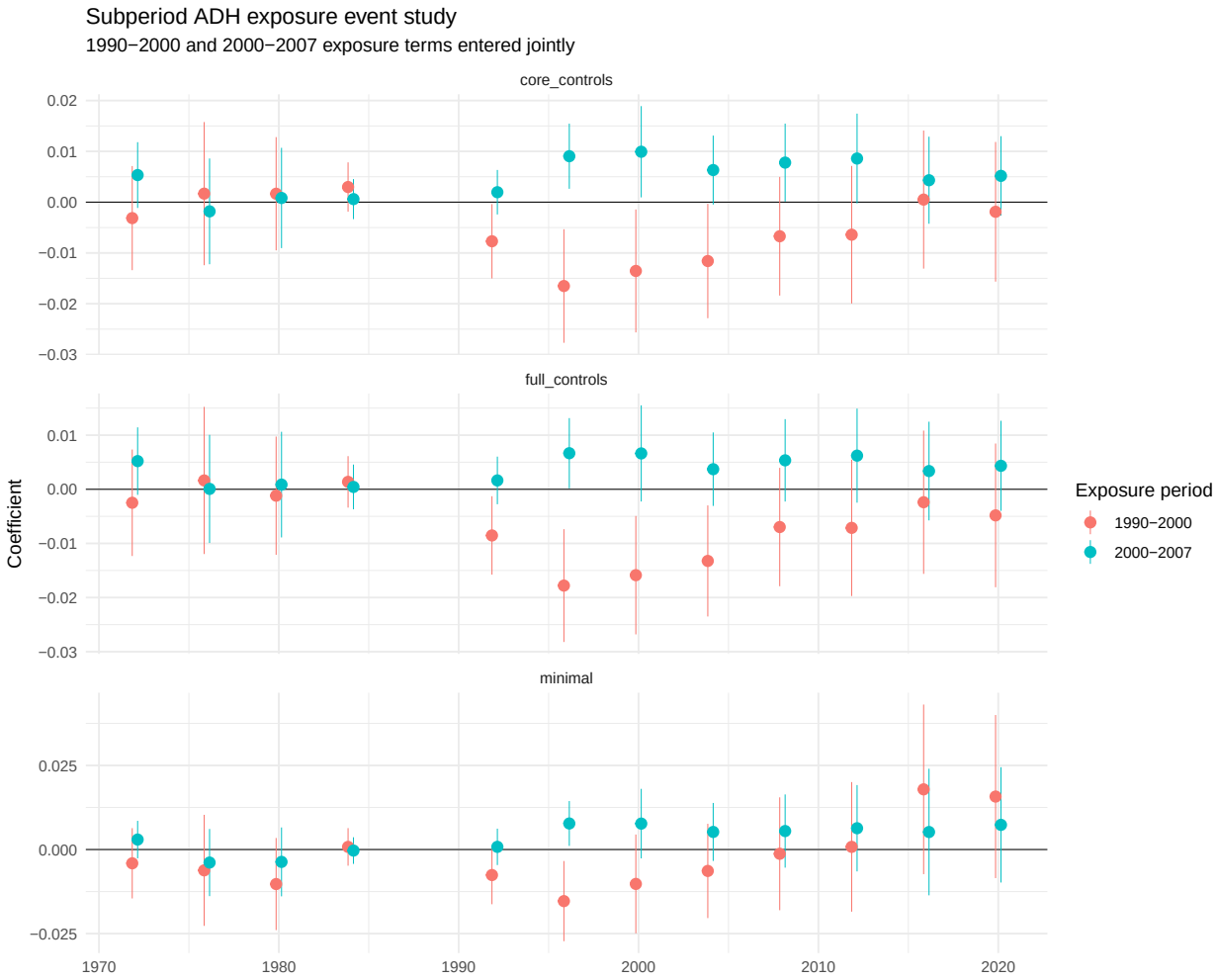


Figure 3: Subperiod ADH exposure event study, with 1990–2000 and 2000–2007 exposure terms entered jointly.

5 Conclusion

In the minimally controlled models, the estimated relationship between Chinese import exposure and Republican presidential margins becomes more positive in 2016 and 2020 relative to 1988. The estimates approach zero with core controls and become slightly negative with full controls. We therefore find no robust positive association across specifications.

The differences across specifications suggest that initial industrial and demographic characteristics are important when interpreting the relationship. Because our political analysis uses observed exposure rather than the instrumental-variables approach of ADH's labor-market study, we cannot interpret these associations as causal effects. Aggregate election results also cannot establish whether residents changed their preferences, whether electoral participation shifted, or whether migration changed the composition of the electorate.

Future research using individual voting data, migration records, and measures of political mobilization could help distinguish these explanations.

6 References

Amlani, Sharif, and Carlos Algara. 2021. "Partisanship and Nationalization in American Elections: Evidence from Presidential, Senatorial, and Gubernatorial Elections in the u.s. Counties, 1872–2020." *Electoral Studies* 73: 102387. https://doi.org/10.1016/j.electstud.2021.102387.

Autor, David H., and David Dorn. 2013. "The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market." *American Economic Review* 103 (5): 1553–97. https://doi.org/10.1257/aer.103.5.1553.

Autor, David H., David Dorn, and Gordon H. Hanson. 2013. "The China Syndrome: Local Labor Market Effects of Import Competition in the United States." *American Economic Review* 103 (6): 2121–68. https://doi.org/10.1257/aer.103.6.2121.

Autor, David H., David Dorn, Gordon H. Hanson, and Kaveh Majlesi. 2020. "Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure." *American Economic Review* 110 (10): 3139–83. https://doi.org/10.1257/aer.20170011.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2025. "Design-Based Identification with Formula Instruments: A Review." *Journal of Economic Perspectives* 39 (2): 157–80. https://doi.org/10.1257/jep.20231370.

Ferrara, Andreas, Patrick A. Testa, and Liyang Zhou. 2024. "Crosswalks for u.s. Counties and Congressional Districts, 1790–2020." *Historical Methods: A Journal of Quantitative and Interdisciplinary History*, ahead of print. https://doi.org/10.1080/01615440.2024.2369230.

Foote, Andrew, Mark Kutzbach, and Lars Vilhuber. 2021. "Recalculating...: How Uncertainty in Local Labour Market Definitions Affects Empirical Findings." *Applied Economics* 53 (14): 1598–612. https://doi.org/10.1080/00036846.2020.1841083.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik Instruments: What, When, Why, and How." *American Economic Review* 110 (8): 2586–624. https://doi.org/10.1257/aer.20181047.

Tolbert, Charles M., and Molly Sizer. 1996. *U.s. Commuting Zones and Labor Market Areas: A 1990 Update*. Economic Research Service, U.S. Department of Agriculture.

U.S. Department of Agriculture Economic Research Service. 2026. *Commuting Zones and Labor Market Areas*. Data product.

A Appendix A: County-to-CZ Harmonization and Measurement

We convert historical county election returns to 1990 county boundaries using the crosswalks of Ferrara et al. (2024), then aggregate votes to the 1990 commuting zones defined by Tolbert and Sizer (1996). We use M5 weights where available and M2 weights otherwise. The following tables summarize the geographic conversion and the proportion of votes retained.

Table 4: Validation checks for the analysis.

Check	Status	Fatal	Count
adh one row per czone	pass	TRUE	0
cz lookup unique 1990 counties	pass	TRUE	0
aa all county fips valid	pass	TRUE	0
aa vote identity error within tolerance	pass	TRUE	0
crosswalk positive final weight sums equal one	pass	TRUE	0
crosswalk all source weight sums equal one or reported	reported	FALSE	0
crosswalk undefined selected weights handled	pass	TRUE	211
aa no duplicate county years after exact distinct	pass	TRUE	0
aa two party votes match party votes	pass	TRUE	0
aa votes nonnegative	pass	TRUE	0
bridged vote retention above threshold	pass	TRUE	0
main panel balanced	pass	TRUE	0
event study residualized design full rank	pass	TRUE	0
event study main vcov available without repair	pass	TRUE	0
event study conley repair diagnostics recorded	pass	FALSE	31

Table 5: Retained vote shares by year and source scope.

Year	Source counties	Fallback counties	Unmatched counties	Retained vote share
1972	3106	258	0	1.0000
1976	3109	258	0	1.0000
1980	3108	221	0	1.0000
1984	3110	221	0	1.0000
1988	3110	221	0	1.0000
1992	3110	0	0	1.0000
1996	3109	0	0	1.0000
2000	3109	196	0	1.0000
2004	3109	196	0	1.0000
2008	3109	196	0	1.0000
2012	3109	183	0	1.0000
2016	3108	183	0	1.0000
2020	3108	184	0	1.0000

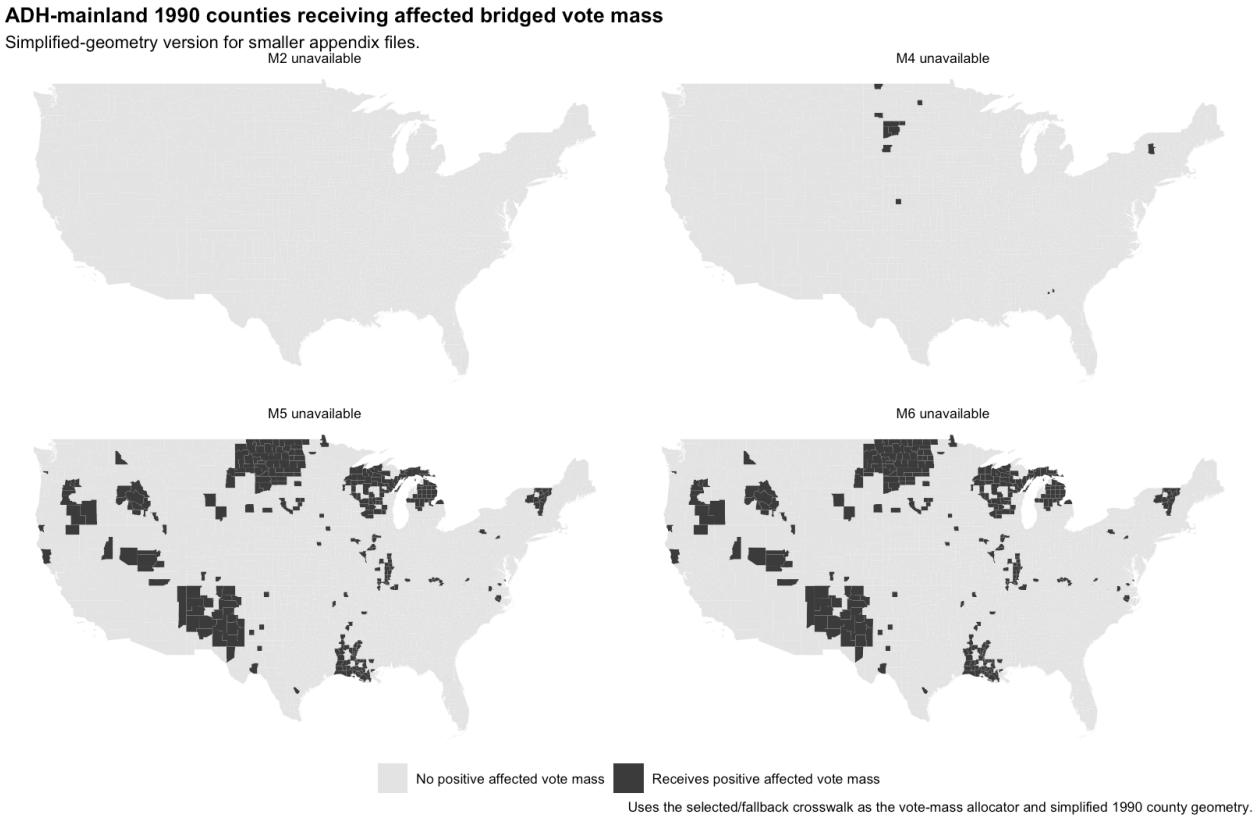


Figure 4: 1990 counties receiving positive vote mass from source counties with unavailable weights.

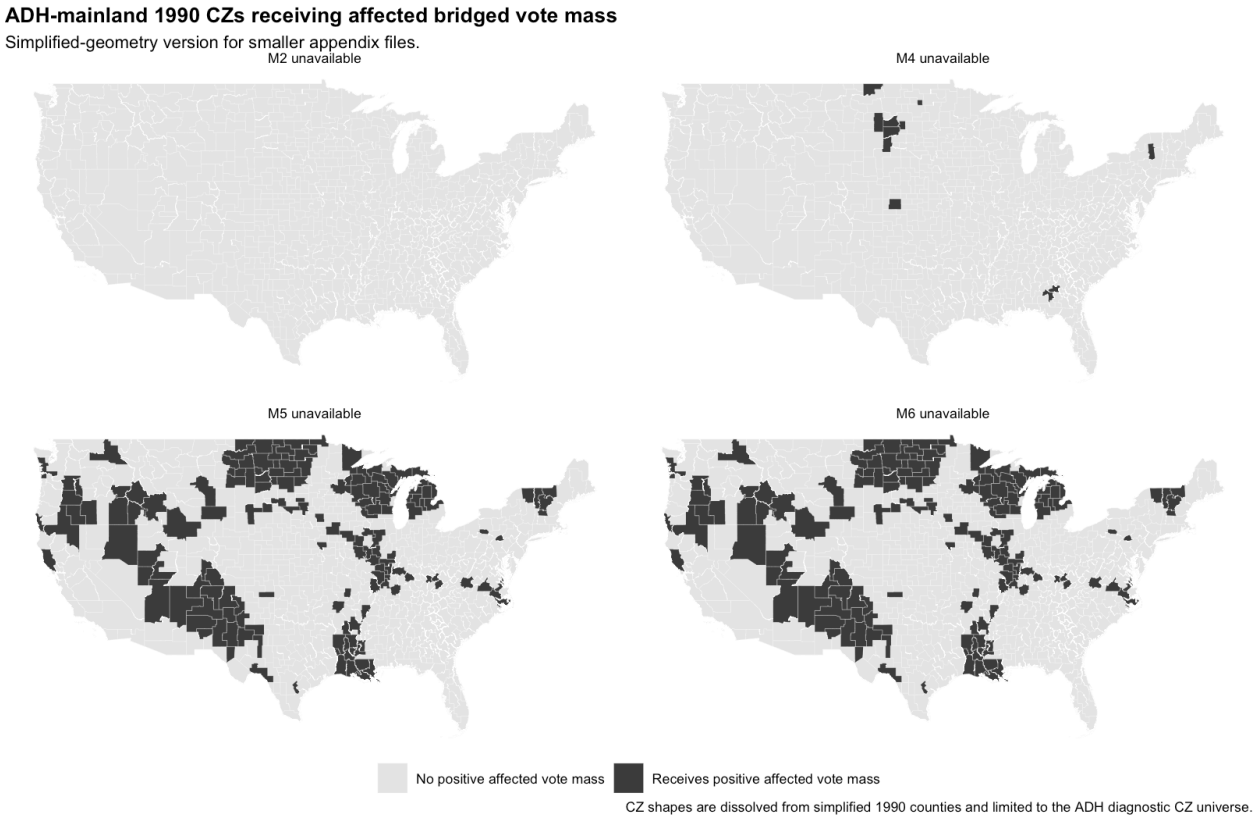


Figure 5: 1990 CZs receiving positive vote mass from source counties with unavailable weights.

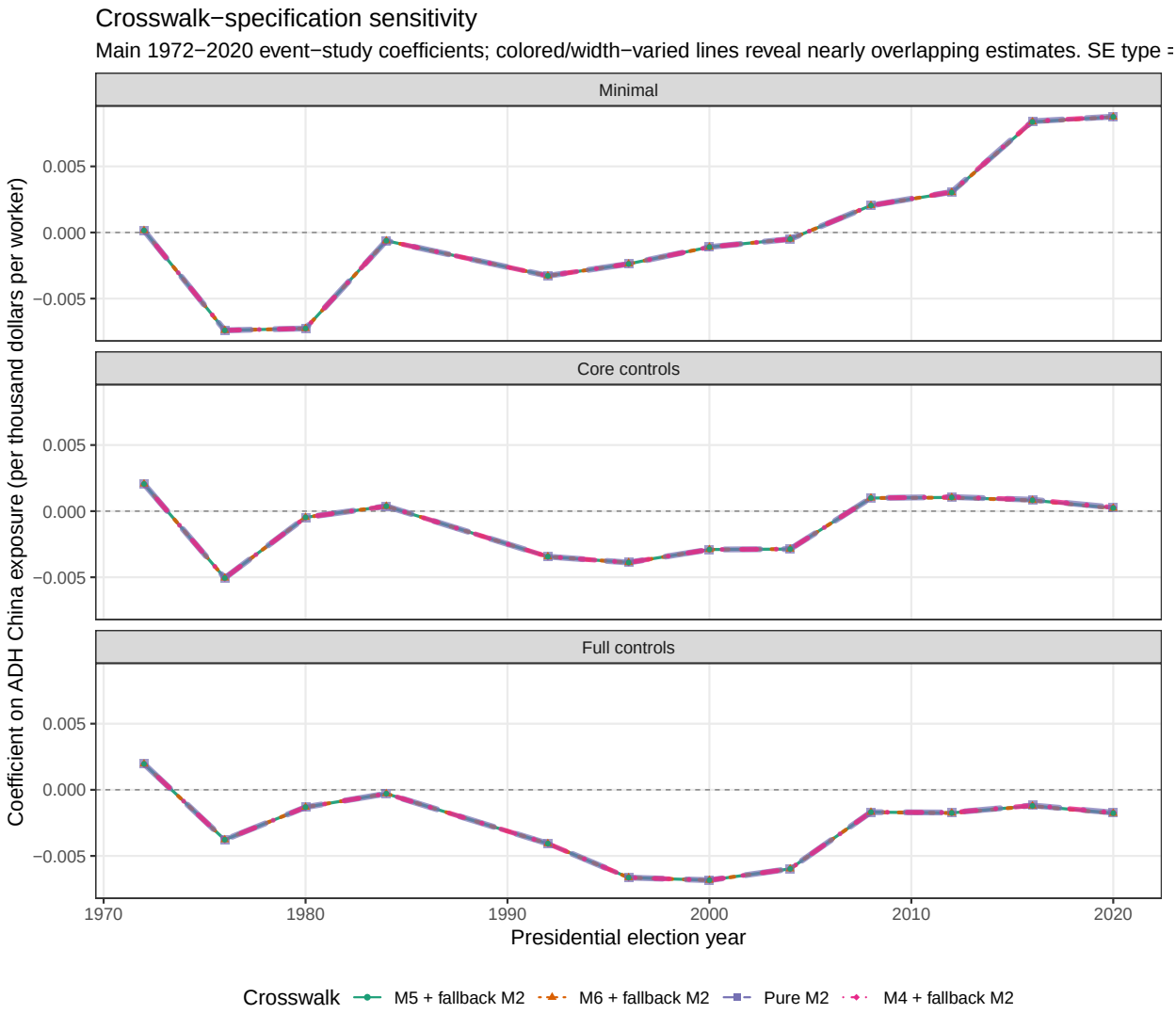


Figure 6: Event-study estimates using alternative county crosswalk weights.

B Appendix B: Main Event Study and Alternative Political Outcomes

Table 6: Main event-study coefficient table, selected years.

Spec	Year	Estimate	Std. error	95% CI	p-value
Minimal	1972	0.0002	0.0019	[-0.0036, 0.0039]	0.936
Minimal	1992	-0.0033	0.0012	[-0.0057, -0.0008]	0.009
Minimal	2000	-0.0011	0.0031	[-0.0073, 0.0051]	0.726
Minimal	2008	0.0020	0.0041	[-0.0061, 0.0101]	0.621
Minimal	2016	0.0084	0.0069	[-0.0051, 0.0218]	0.224
Minimal	2020	0.0087	0.0062	[-0.0034, 0.0209]	0.159
Core controls	1972	0.0021	0.0025	[-0.0029, 0.0070]	0.412

Table 6: Main event-study coefficient table, selected years.

Spec	Year	Estimate	Std. error	95% CI	p-value
Core controls	1992	-0.0034	0.0013	[-0.0061, -0.0008]	0.011
Core controls	2000	-0.0029	0.0029	[-0.0085, 0.0027]	0.313
Core controls	2008	0.0010	0.0027	[-0.0043, 0.0063]	0.713
Core controls	2016	0.0008	0.0025	[-0.0041, 0.0057]	0.740
Core controls	2020	0.0002	0.0027	[-0.0050, 0.0055]	0.930
Full controls	1972	0.0020	0.0024	[-0.0028, 0.0067]	0.419
Full controls	1992	-0.0041	0.0014	[-0.0068, -0.0013]	0.004
Full controls	2000	-0.0068	0.0028	[-0.0123, -0.0013]	0.015
Full controls	2008	-0.0017	0.0030	[-0.0075, 0.0041]	0.569
Full controls	2016	-0.0012	0.0028	[-0.0068, 0.0043]	0.668
Full controls	2020	-0.0018	0.0029	[-0.0074, 0.0038]	0.537

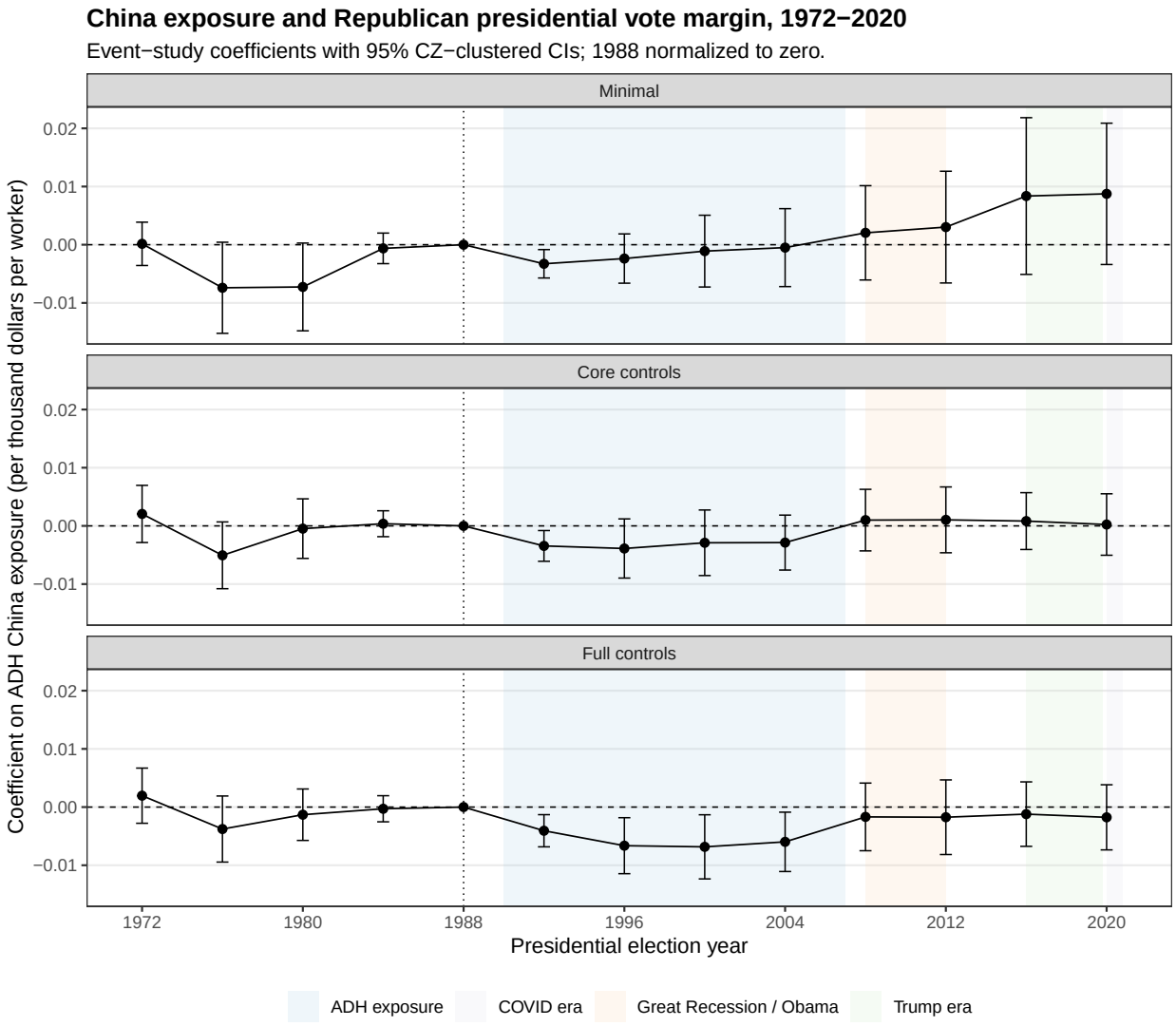


Figure 7: Main event-study estimates.

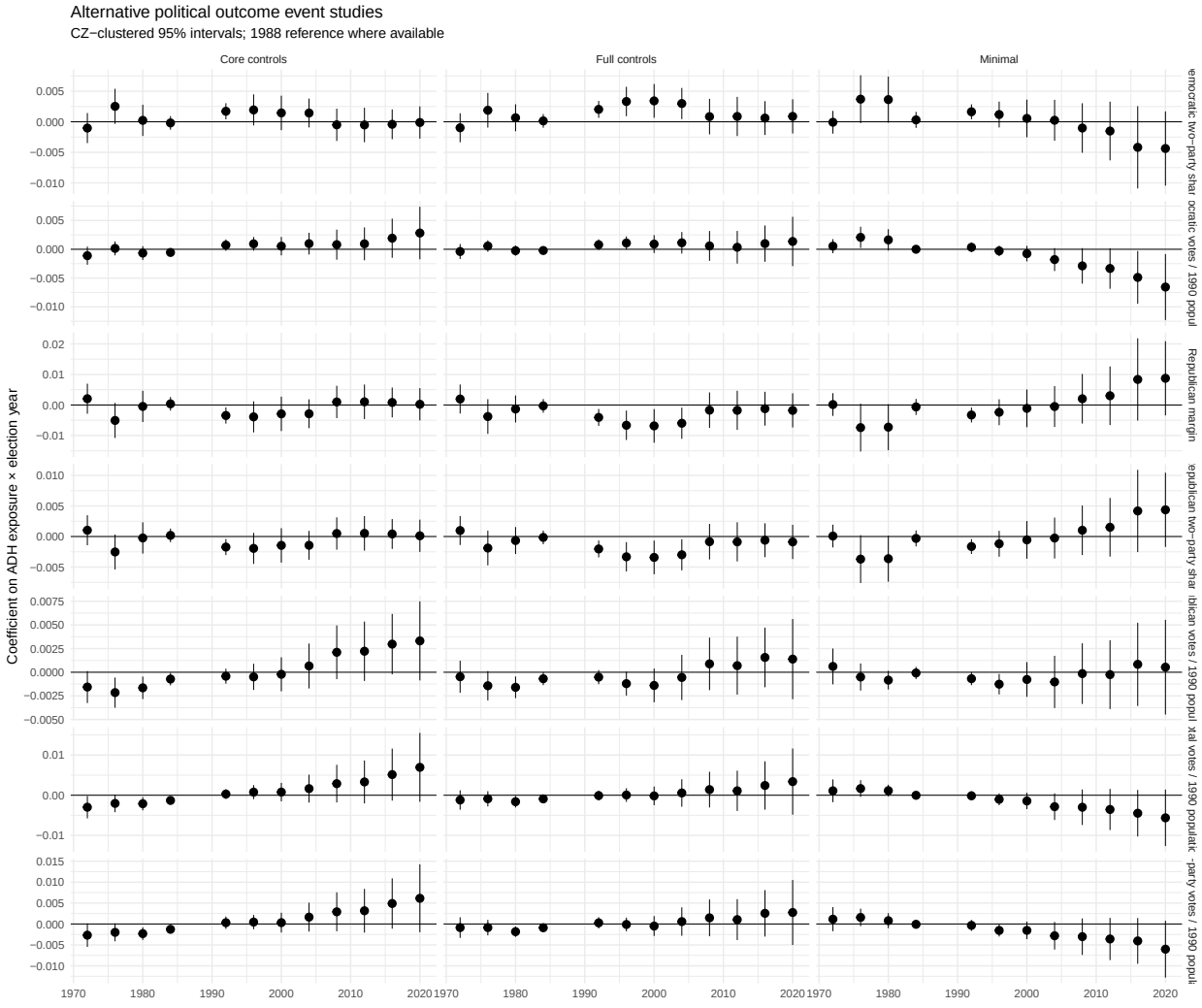


Figure 8: Full alternative-outcome event-study grid.

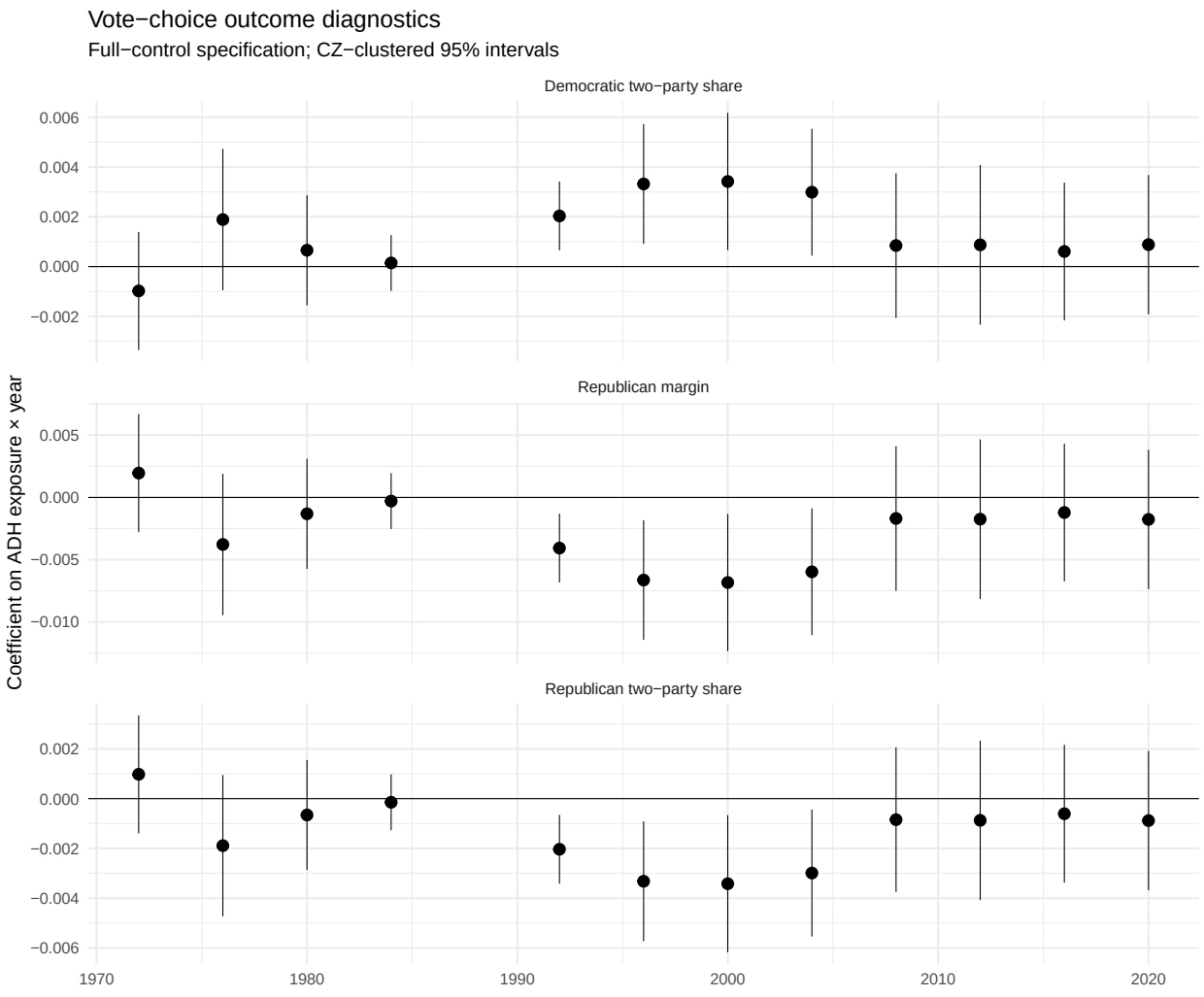


Figure 9: Vote-choice outcome family.

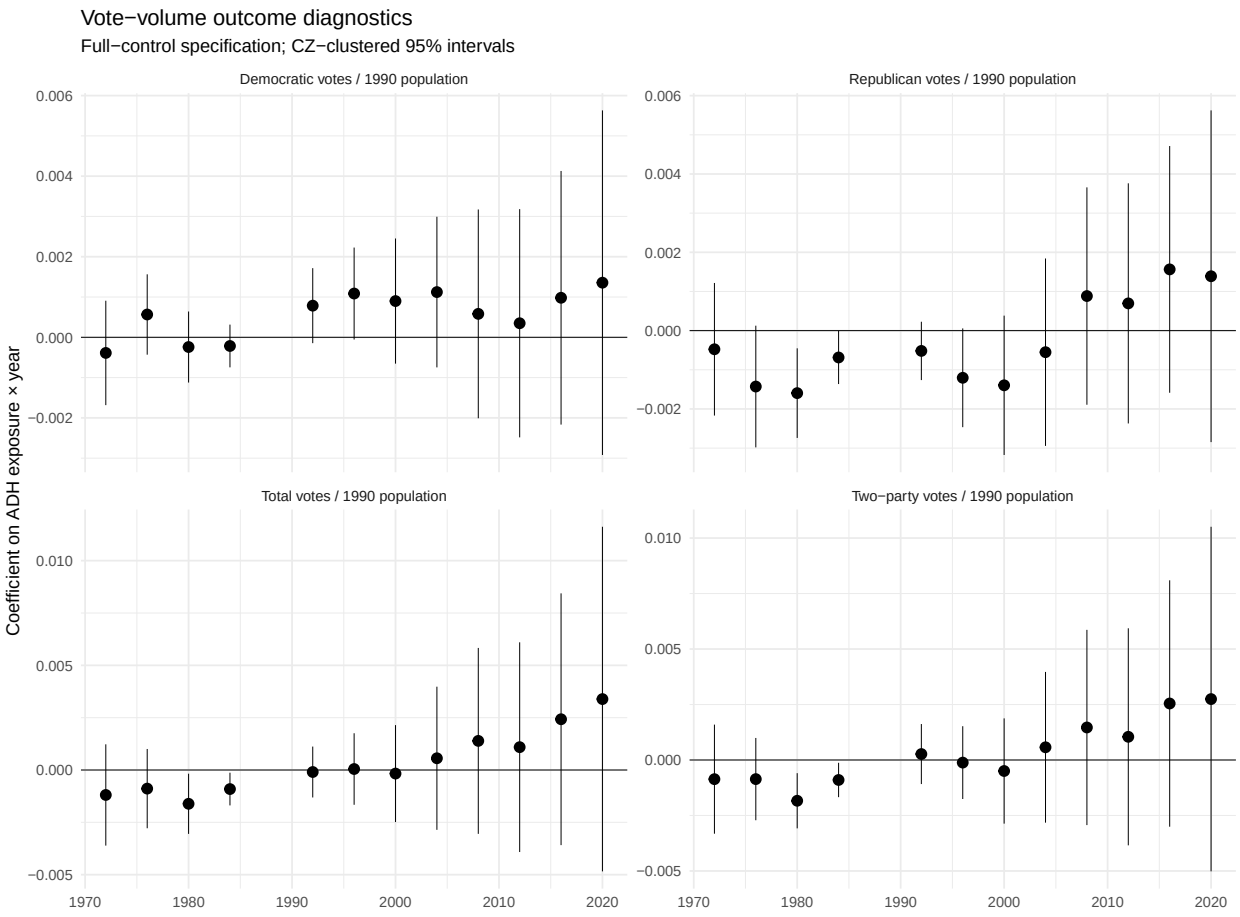


Figure 10: Vote-volume outcome family.

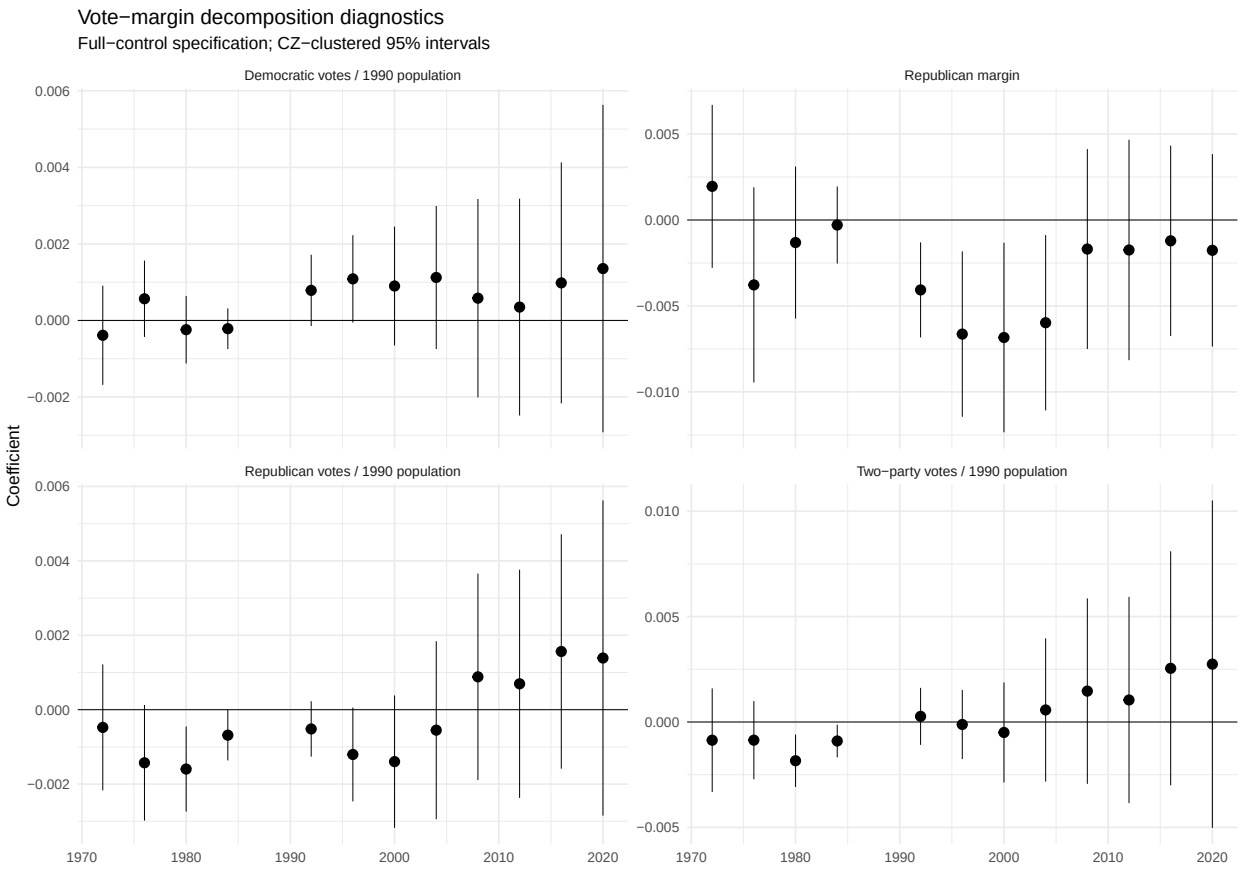


Figure 11: Outcome decomposition.

C Appendix C: Bartik Identification Diagnostics

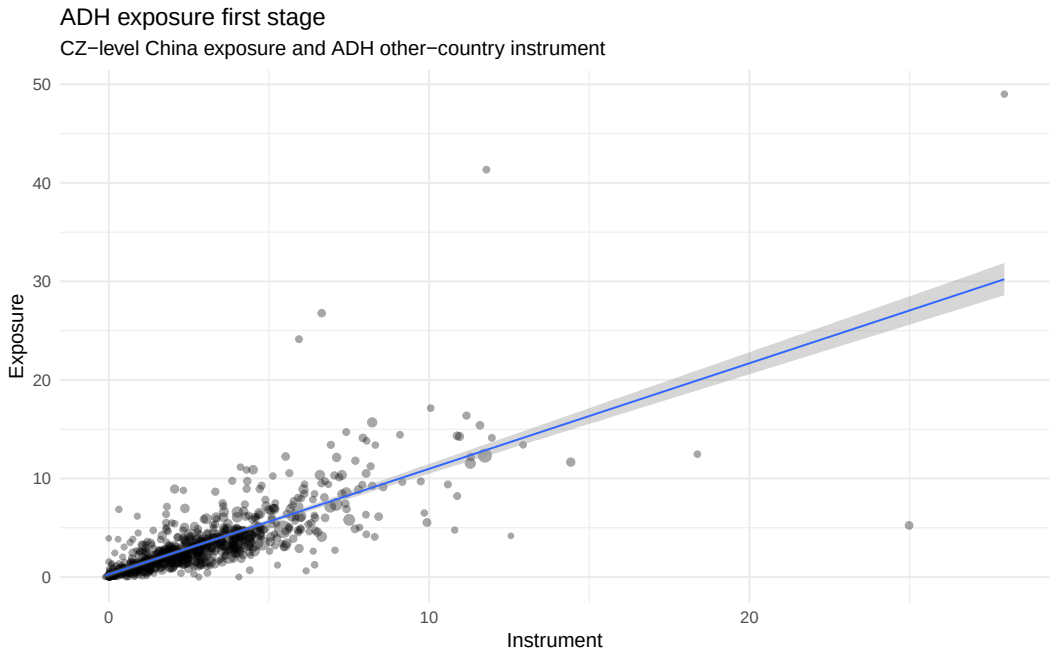


Figure 12: First-stage relationship between ADH exposure and the ADH instrument.

Table 7: Baseline balance: correlations with exposure and instrument.

Variable	Corr. with exposure	Corr. with instrument	Weighted mean	SD
l shind manuf cbp	0.574	0.614	20.695	12.848
l sh popedu c	-0.243	-0.287	48.021	8.574
l sh popfborn	-0.145	-0.131	10.146	4.969
l sh empl f	0.028	0.006	63.624	6.634
l sh routine33	0.301	0.355	32.228	3.362
l task outsource	0.261	0.312	0.112	0.407
statefip	-0.085	-0.087	27.790	14.987

Table 8: Pre-1988 Republican-margin placebo diagnostics.

Outcome	Spec	Regressor	Estimate	Std. error	p-value
pretrend slope rep margin	Minimal	exposure 1990 2007	0.0001	0.0002	0.601
pretrend slope rep margin	Core controls	exposure 1990 2007	-0.0001	0.0002	0.700
pretrend slope rep margin	Full controls	exposure 1990 2007	-0.0001	0.0002	0.548
pretrend slope rep margin	Minimal	instrument 1990 2007	0.0000	0.0002	0.976

Table 8: Pre-1988 Republican-margin placebo diagnostics.

Outcome	Spec	Regressor	Estimate	Std. error	p-value
pretrend slope rep margin	Core controls	instrument 1990 2007	-0.0001	0.0002	0.556
pretrend slope rep margin	Full controls	instrument 1990 2007	-0.0002	0.0002	0.498
pretrend change first to last	Minimal	exposure 1990 2007	0.0075	0.0062	0.236
pretrend change first to last	Core controls	exposure 1990 2007	-0.0012	0.0056	0.833
pretrend change first to last	Full controls	exposure 1990 2007	-0.0018	0.0061	0.769
pretrend change first to last	Minimal	instrument 1990 2007	0.0064	0.0083	0.445
pretrend change first to last	Core controls	instrument 1990 2007	-0.0001	0.0081	0.990
pretrend change first to last	Full controls	instrument 1990 2007	-0.0004	0.0087	0.959

Table 9: Industry-shift summary from ADH trade-data diagnostics.

Status	Period	Importer	SIC87DD	Start imports	End imports	Import growth
ok	1991-1999	OTH	3679	143699.7	2505878.0	2362178.3
ok	1991-1999	OTH	3577	40964.0	1565398.4	1524434.3
ok	1991-1999	OTH	3944	1285590.8	2732429.2	1446838.5
ok	1991-1999	OTH	3651	936563.7	2377214.8	1440651.1
ok	1991-1999	OTH	3861	58608.0	1004058.8	945450.8
ok	1991-1999	OTH	2369	1191708.4	2126562.8	934854.4
ok	1991-1999	OTH	2337	1214870.9	2095917.0	881046.1
ok	1991-1999	OTH	2091	235925.2	1114860.0	878934.8
ok	1991-1999	OTH	3161	623382.1	1466893.6	843511.5
ok	1991-1999	OTH	3089	256498.7	1082972.6	826474.0
ok	1991-1999	OTH	2325	634508.3	1300735.5	666227.2
ok	1991-1999	OTH	2392	605945.9	1227629.4	621683.4

D Appendix D: Subperiod Exposure and Dynamic Interpretation

We report differences between the early- and late-exposure coefficients descriptively. Formal inference would require accounting for their covariance.

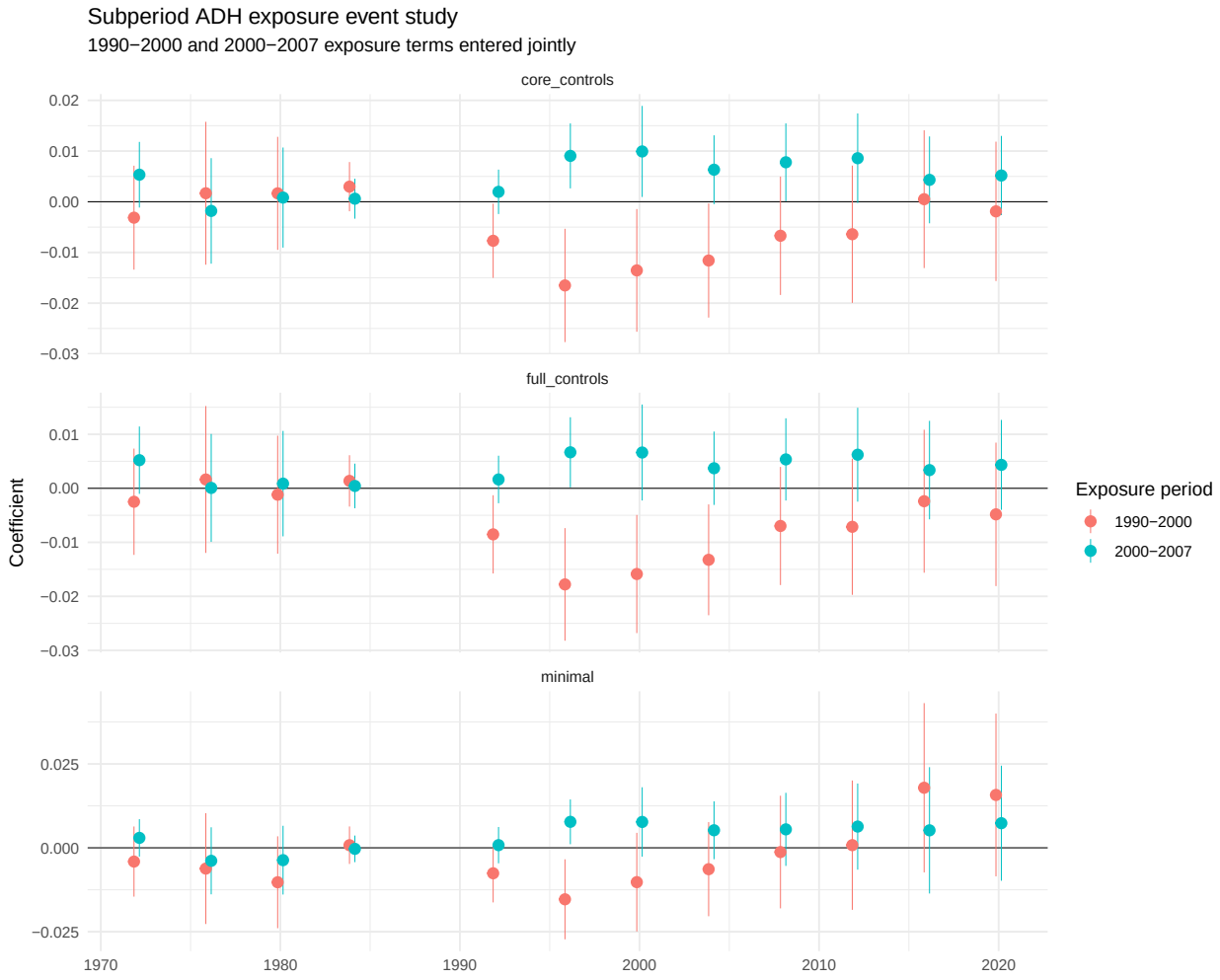


Figure 13: Subperiod exposure event-study estimates.

Table 10: Selected horizons: early versus late ADH exposure.

Spec	Year	1990–2000 exposure	2000–2007 exposure	Early minus late
Minimal	1996	-0.0153	0.0077	-0.0231
Minimal	2000	-0.0102	0.0077	-0.0179
Minimal	2008	-0.0013	0.0055	-0.0068
Minimal	2016	0.0179	0.0052	0.0127
Minimal	2020	0.0157	0.0073	0.0084
Core controls	1996	-0.0165	0.0090	-0.0256
Core controls	2000	-0.0135	0.0099	-0.0235
Core controls	2008	-0.0067	0.0078	-0.0145
Core controls	2016	0.0005	0.0043	-0.0038
Core controls	2020	-0.0019	0.0052	-0.0071
Full controls	1996	-0.0178	0.0067	-0.0245
Full controls	2000	-0.0159	0.0066	-0.0225
Full controls	2008	-0.0070	0.0053	-0.0123

Table 10: Selected horizons: early versus late ADH exposure.

Spec	Year	1990–2000 exposure	2000–2007 exposure	Early minus late
Full controls	2016	-0.0024	0.0034	-0.0058
Full controls	2020	-0.0048	0.0043	-0.0092

E Appendix E: Standard-Error Diagnostics

We cluster the main standard errors by commuting zone. The following tables summarize alternative variance estimators and potential numerical problems.

Table 11: Variance-covariance diagnostics by SE type.

SE type	status	Model rows	Repairs needed	Max negative eigenvalues	Smallest eigenvalue
CZ x year	repair_required	8	8	94	-0.004036
CZ-clustered	ok	8	0	0	0.000000
Conley 1000 km	repair_required	8	8	57	-0.005521
Conley 250 km	ok	1	0	0	0.000000
Conley 250 km	repair_required	7	7	39	-0.000221
Conley 500 km	repair_required	8	8	54	-0.001218
Conley 750 km	repair_required	8	8	55	-0.002513
Driscoll-Kraay	ok	8	0	51	-0.000000
HC	ok	8	0	0	0.000000
Newey-West	ok	8	0	0	0.000000
State-clustered	ok	8	0	35	-0.000000

Table 12: SE diagnostic issue summary.

SE type	Issue	Model rows	Max near-zero SEs	Max nonfinite SEs	Max rank deficiency
CZ x year	Two-way cluster: few year clusters	8	0	10	0
Conley 1000 km	Conley VCOV non-PD or repaired	8	0	1	0
Conley 250 km	Conley VCOV non-PD or repaired	7	0	0	0

Table 12: SE diagnostic issue summary.

SE type	Issue	Model rows	Max near-zero SEs	Max nonfinite SEs	Max rank deficiency
Conley 500 km	Conley VCOV non-PD or repaired	8	0	0	0
Conley 750 km	Conley VCOV non-PD or repaired	8	0	1	0
Driscoll-Kraay	Near-zero or nonfinite SEs	8	17	0	80
State-clustered	Rank-deficient VCOV	5	0	0	72

F Appendix F: Additional Political Outcomes

F.1 NaNDA turnout, vote ratios, and partisanship

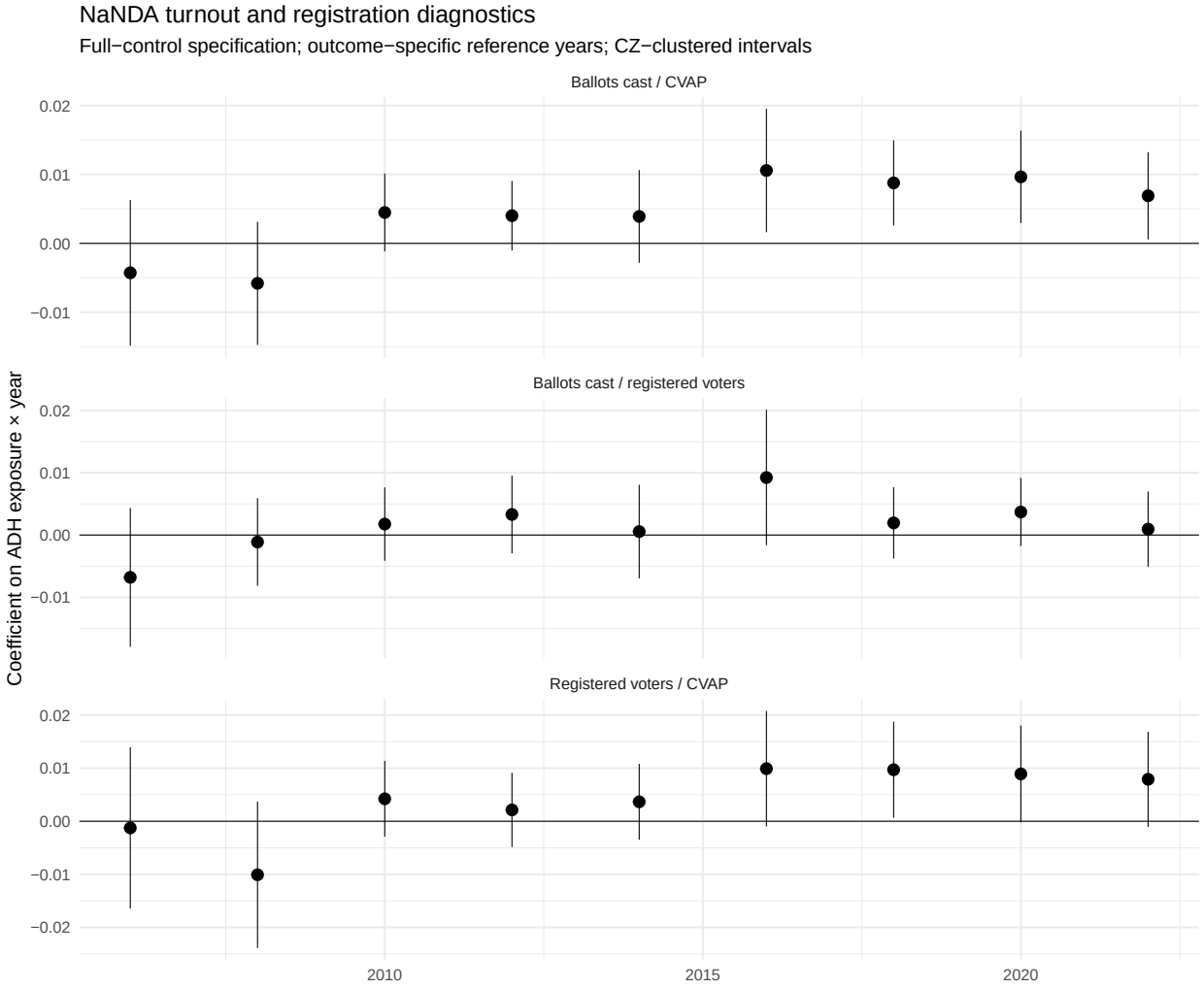


Figure 14: NaNDA turnout and registration outcomes.

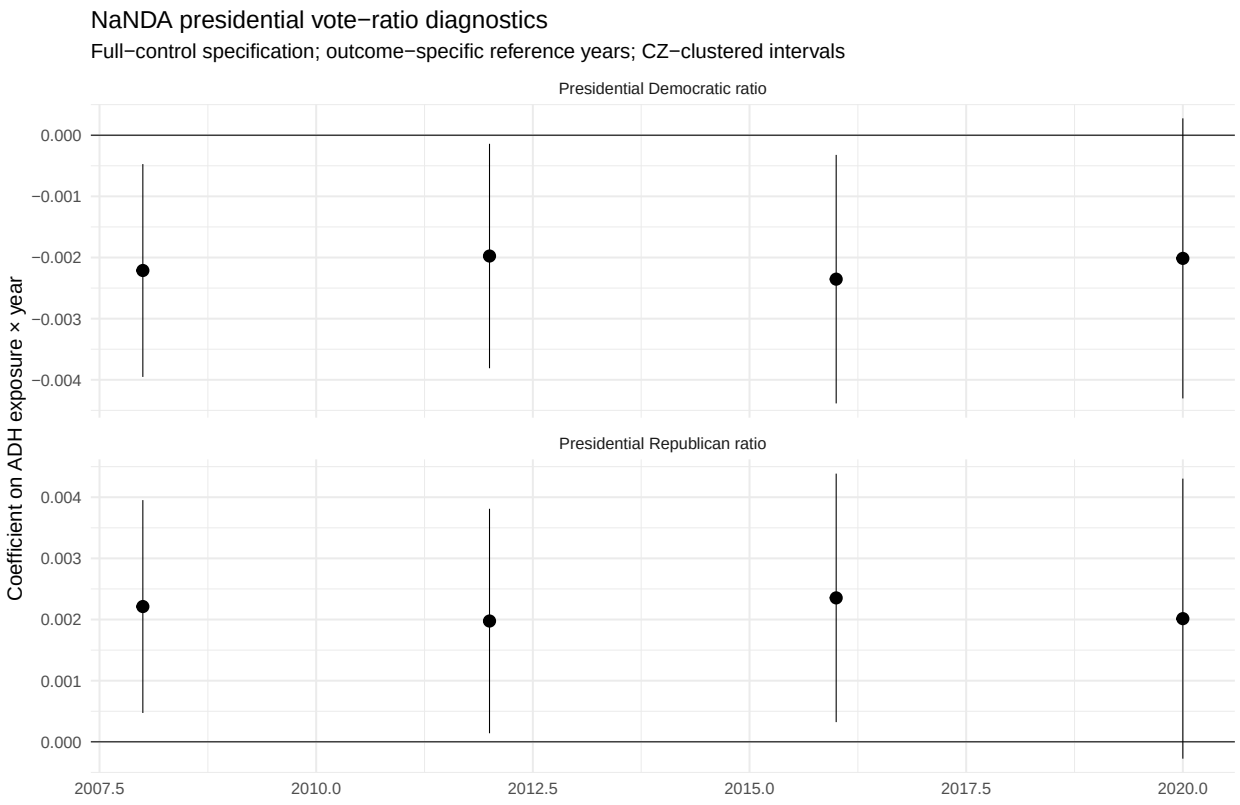


Figure 15: NaNDA presidential vote ratios.

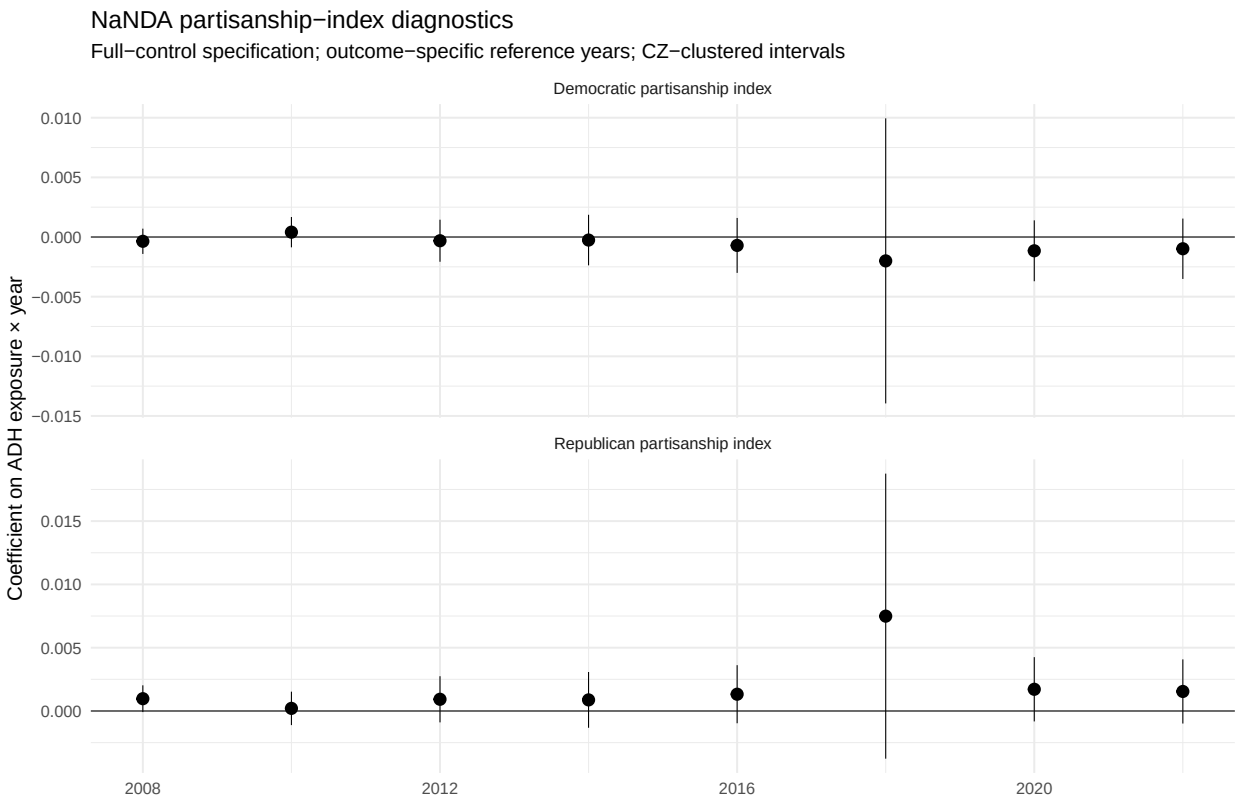


Figure 16: NaNDA partisanship indices.

Table 13: NaNDA outcome availability summary.

Outcome	Valid years	Unavailable years	Min non-missing CZs	Years with all-zero values
Ballots / CVAP	2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	None	721	None
Ballots / registered voters	2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	None	661	None
Democratic partisanship index	2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	2004	722	2004
Presidential Democratic ratio	2004, 2008, 2012, 2016, 2020	2006, 2010, 2014, 2018, 2022	722	None
Presidential Republican ratio	2004, 2008, 2012, 2016, 2020	2006, 2010, 2014, 2018, 2022	722	None
Registered / CVAP	2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	None	720	None

Table 13: NaNDA outcome availability summary.

Outcome	Valid years	Unavailable years	Min non-missing CZs	Years with all-zero values
Republican partisanship index	2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	2004	722	2004

F.2 Political Representation and Related Outcomes

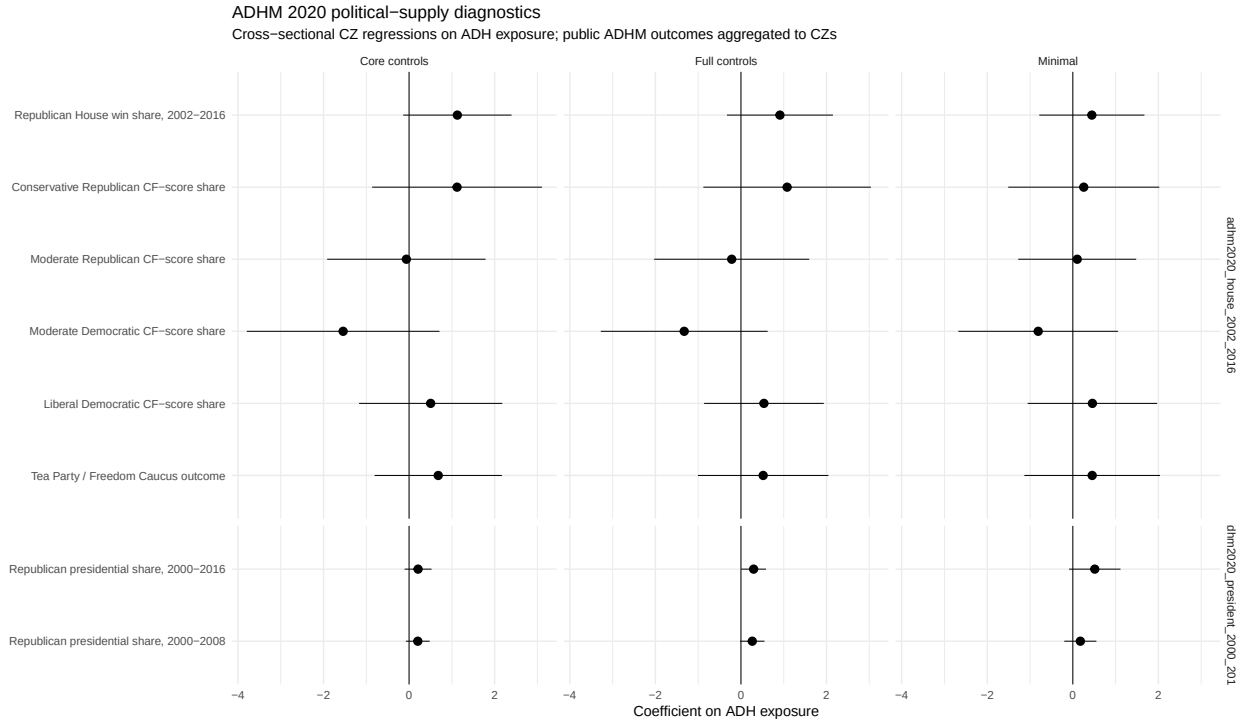


Figure 17: Political representation and related outcomes from ADHM (2020).

Table 14: ADHM (2020) results, full-control specification.

Outcome	Estimate	Std. error	p-value	N
Republican House win share, 2002-2016	0.911	0.632	0.156	722
Conservative Republican CF-score share	1.080	0.999	0.285	722
Moderate Republican CF-score share	-0.216	0.925	0.816	722
Moderate Democratic CF-score share	-1.326	0.995	0.189	722
Liberal Democratic CF-score share	0.538	0.714	0.455	722
Tea Party / Freedom Caucus outcome	0.520	0.778	0.508	722
Republican presidential share, 2000-2016	0.296	0.148	0.052	722
Republican presidential share, 2000-2008	0.265	0.145	0.075	722

G Appendix G: Replication Notes

The complete analysis can be regenerated from the project root with:

```
source(here::here("src", "run_pipeline.R"))
```

To reproduce estimates under all four county crosswalk specifications:

```
Sys.setenv(RUN_CROSSWALK_SENSITIVITY = "true")
source(here::here("src", "run_pipeline.R"))
```

The replication-code PDF can be generated from the `report` folder with:

```
source("make_replication_code_pdf.R")
```